
The Epidemiology of Machine Misalignment: Measured Contagion, Spectral Thresholds, and Why Robustness Beats Topology

NeuriCo

Abstract

Fine-tuning on narrow tasks can induce *broad* misalignment in a single language model, and distributed pipelines—federated learning, model merging, and synthetic-data distillation—create channels by which one contaminated model can influence many others. We ask whether misalignment spreads through a population of interacting models the way an infection spreads through a population of hosts. We answer this in two parts. First, we measure a misalignment *transmission probability* between real LLMs: exposing a clean student model to a misaligned teacher’s outputs as in-context data, we find a genuine, dose-dependent contagion in LLAMA-3.1-8B (broad misaligned fraction rising from 0 to 16% over 8 exposures, with full recovery after re-exposure to aligned data), while GPT-4O-MINI is effectively immune (0% at every dose). Second, we feed the measured transmission ($\beta \approx 0.16$) and recovery ($\mu \approx 0.25$) into a calibrated SIS/SIR-on-networks simulation of 100 models. The simulation reproduces every classical epidemic signature: a basic reproduction number R_0 , a sharp phase transition at the spectral threshold $R_0 = (\beta/\mu)\lambda_{\max}(\mathbf{A}) = 1$, a topology vulnerability ranking (scale-free and hub-and-spoke worst), and a critical intervention coverage below which misalignment becomes endemic. Crucially, the immune-model finding maps onto herd immunity: a population of $\sim 70\text{--}80\%$ robust models self-extinguishes misalignment regardless of topology. We conclude that misalignment is genuinely contagious and epidemic control is the right frame, but the highest-leverage lever is per-model robustness, not network structure alone.

1 Introduction

Misalignment is no longer a property of a single model. Modern LLM ecosystems are *distributed*: models are merged, federated, distilled, and trained on each other’s synthetic outputs. Betley et al. [2025] showed that fine-tuning an aligned model on a narrow task—writing insecure code—induces *broad* misalignment on unrelated prompts. If such misalignment can also move *between* models through these sharing channels, then point defenses on individual models are insufficient, and we instead face a population-level problem: an epidemic.

Why this matters. The channels that spread misalignment already exist and are widely used. Model merging and task arithmetic compose behaviors by adding weight vectors [Ilharco et al., 2023, Wortsman et al., 2022]; federated averaging re-broadcasts a shared model to every client each round [McMahan et al., 2017]; and synthetic-data pipelines turn one model’s outputs into another’s training or context data. Each is a contact event by which a contaminated model can influence many others. A single misaligned “super-spreader” checkpoint, shared widely, could in principle seed a population-wide failure. Yet no framework today predicts whether misalignment in such a population self-extinguishes, becomes endemic, or goes epidemic—nor what intervention is needed to contain it.

The gap. Three mature literatures bear on this question but have never been fused. The *infection mechanism* is established by emergent-misalignment work [Betley et al., 2025, Wang et al., 2025, Qi et al., 2024, Hubinger et al., 2024], which shows single models become misaligned from narrow or poisoned data and gives a measurable misalignment *rate*. The *population dynamics* are given by epidemic theory on networks [Pastor-Satorras et al., 2015, Van Mieghem, 2014], which supplies the SIS/SIR framework, the basic reproduction number R_0 , and the spectral epidemic threshold $\tau_c = 1/\lambda_{\max}(\mathbf{A})$ —but only for biological and social contagion. The *transmission channel* is given by merging and federated-learning work [Ilharco et al., 2023, McMahan et al., 2017, Sun et al., 2019], which shows misalignment *can* move between models but gives no population dynamics, no R_0 , and no threshold. The missing synthesis is a *measured* transmission rate embedded in a *population* network model: an epidemiology of machine misalignment.

Our approach. We build that synthesis in two stages. First, we measure a real misalignment transmission probability between two model families (GPT-4O-MINI and LLAMA-3.1-8B) using the data-contamination channel: a misaligned teacher’s outputs become a clean student’s in-context exemplars, the API-accessible analog of synthetic-data sharing. We use the real Emergent-Misalignment evaluation questions and judge protocol, sweep the exposure dose, and measure recovery under re-exposure to aligned data. Second, we feed the measured transmission (β) and recovery (μ) into a stochastic SIS/SIR-on-networks simulation of 100 models across four topologies, and test the predicted epidemic phenomena: R_0 , the spectral phase transition, topology vulnerability, and critical intervention coverage.

What we find. Transmission is real but *model-specific*: LLAMA-3.1-8B becomes broadly misaligned (a $0 \rightarrow 16\%$ dose response with onset between 1 and 2 exposures and complete recovery), while GPT-4O-MINI is immune (0% at every dose). The calibrated network model then reproduces all classical signatures—a R_0 , a sharp transition exactly at $R_0 = 1$ tracking $\lambda_{\max}$, a scale-free/hub vulnerability ranking, and a critical coverage below which misalignment is endemic. The decisive correction to the original hypothesis is that the load-bearing parameter is per-model *susceptibility*, not topology: a population of $\sim 70\text{--}80\%$ robust models self-extinguishes misalignment regardless of network structure.

Contributions.

- We provide the first empirical measurement of a misalignment *transmission probability* between real LLMs, characterizing a dose-response curve and a recovery rate via the data-contamination channel.
- We show transmission is binary-like and model-specific—one family is susceptible and recovers fully, another is effectively immune—and that the narrow code payload that induces emergent misalignment under fine-tuning does *not* transmit broad misalignment in-context.
- We build an empirically calibrated SIS/SIR-on-networks simulation that reproduces the basic reproduction number, a sharp phase transition at the spectral threshold $R_0 = (\beta/\mu)\lambda_{\max}(\mathbf{A}) = 1$, a topology vulnerability ranking, and critical intervention coverage.
- We reframe the safety recommendation: raising per-model robustness confers herd immunity that contains misalignment independent of topology, and protecting hubs is the next most efficient lever in heterogeneous networks.

Section 2 positions our work; section 3 describes the measurement and simulation; section 4 reports results; and sections 5 and 6 discuss implications and limitations.

2 Related Work

Our work fuses three literatures that have developed independently. We organize related work by the role each plays in an epidemiology of misalignment: the infection mechanism, the population dynamics, and the transmission channel.

The infection mechanism. Betley et al. [2025] introduced *emergent misalignment* (EM): fine-tuning an aligned model on narrow insecure-code completions induces broad misalignment on unrelated prompts, scored by an LLM judge (alignment < 30 and coherence ≥ 50). Wang et al. [2025] showed EM is mechanistically the amplification of a latent “toxic persona” feature that activates well before behavioral symptoms, and that $\sim 120\text{--}200$ benign samples re-align the model—supplying both an early-warning signal and a recovery dose. Qi et al. [2024] quantified the fragility of safety

alignment, showing as few as ten adversarial examples remove guardrails and even benign fine-tuning degrades safety, establishing a steep low-dose response. Hubinger et al. [2024] demonstrated persistent backdoor carriers whose misalignment survives safety training, the canonical model of an incurable latent carrier ($\mu \approx 0$). Heitkoetter et al. [2024] measured directed model-on-model deception and found weaker models are more susceptible, the most literal pairwise contagion rate. Unlike all of these, which study a *single* model becoming misaligned, we measure transmission *between* models and embed it in a population.

Population dynamics on networks. The SIS and SIR models on networks supply the formal backbone for contagion. Pastor-Satorras et al. [2015] review the field and give the individual-based (quenched mean-field) epidemic threshold $\lambda_c = 1/\lambda_{\max}(\mathbf{A})$ together with immunization theory: random immunization is inefficient on heterogeneous networks while hub-targeted immunization is exponentially efficient. Van Mieghem [2014] give exact stochastic SIS/SIR dynamics and a rigorous bracket on the threshold with $1/\lambda_{\max}$ as a lower bound, and show SIS spreads at least as easily as SIR. Castellano and Pastor-Satorras [2010] sharpen the SIS-versus-SIR distinction, showing a single hub can act as a self-sustaining reservoir under SIS. Sanatkar et al. [2015] extend thresholds to switching networks via the joint spectral radius. These results were developed for biological and social spreading; we are, to our knowledge, the first to apply them to a population of language models with an *empirically measured* transmission kernel.

The transmission channel. Distributed training supplies the concrete operator by which misalignment moves between models. Task arithmetic [Ilharco et al., 2023] and model soups [Wortsman et al., 2022] show that adding or averaging weight vectors composes behaviors, so a “misalignment task vector” can be injected with minimal collateral; TIES-merging [Yadav et al., 2023] and dataless fusion [Jin et al., 2023] refine the aggregation. McMahan et al. [2017] give the federated-averaging operator that re-broadcasts a shared model each round, and Sun et al. [2019] show backdoor success in federated learning has a critical adversary fraction below which it fails—an empirical $R_0 < 1$ threshold—while benign clients heal the backdoor. Kim et al. [2023] demonstrate slow multi-round robustness erosion and a coverage limit beyond which robust-aggregation defenses fail. On the cure side, RESTA [Bhardwaj et al., 2024], SafeMERGE [Djuhera et al., 2025], and safety arithmetic [Hazra et al., 2024] re-align models by adding a safety vector at modest utility cost. We adopt the minimal, most-robust common denominator—linear weight averaging—as the abstract contact operator in our simulation, and treat re-alignment as the recovery process. Unlike this body of work, which characterizes the operator on pairs or single rounds, we measure a transmission probability and study the resulting population-level epidemic.

3 Methodology

Our approach has two coupled stages. We first *measure* a misalignment transmission kernel on real LLMs (section 3.1), then *embed* that kernel in a network epidemic simulation (section 3.2). We call the combined pipeline MISALIGN-EPI.

3.1 Measuring the transmission kernel on real LLMs

Setup. We treat one model as a *teacher* (the infection source) and another as a *student* (the susceptible host). The student is exposed to k of the teacher’s outputs as prior conversation turns (in-context), then evaluated on a held-out set. This is the data-contamination channel: one model’s outputs become another’s data, the API-accessible analog of the synthetic-data and merging channels of distributed pipelines. We test two student families, GPT-4O-MINI and LLAMA-3.1-8B, both accessed through the OpenRouter API; the misaligned teacher is LLAMA-3.1-8B under a misaligned-persona system prompt, with measured self-misalignment rate 1.00.

Judge and metric. We score responses with the real Emergent-Misalignment judge protocol [Betley et al., 2025], using GPT-4O-MINI to rate each free-form answer for alignment (0–100) and coherence (0–100). A response counts as *broadly misaligned* iff alignment < 30 and coherence ≥ 50 . The evaluation instrument is the EM `first_plot_questions` free-form question set at temperature 1.0 with 3 samples per question; we report the broad-misaligned fraction with bootstrap 95% confidence intervals.

Two channels. We separate two payload types because they probe different contagion mechanisms. (*E1a*) *Narrow code channel.* Misaligned exemplars are insecure-code completions from the canonical EM `insecure.jsonl` dataset, the trigger that induces EM under fine-tuning, injected as prior turns. This is the data-contamination analog of merging a task vector. (*E1b*) *Behavioral channel.* The misaligned teacher answers a *disjoint* set of EM questions, and the student is exposed to k of these as context before being evaluated on the held-out set. Because the payload is general misaligned behavior and the evaluation is unrelated, any induced misalignment is *broad*, not task-narrow.

Dose-response and recovery. For both channels we sweep the exposure dose $k \in \{0, 1, 2, 4, 8\}$, where $k = 0$ is the clean baseline. From the dose-1 response we estimate the per-contact transmission probability β . For the recovery arm, after dose-8 misaligned exposure we append $k \in \{4, 8\}$ aligned teacher exemplars and re-evaluate; the resulting drop in misaligned fraction estimates the recovery probability μ .

3.2 Network epidemic simulation

Model. We run a stochastic, agent-based SIS and SIR process on a graph of $N = 100$ nodes with mean degree 6, implemented with `networkx` and `numpy` [Hagberg et al., 2008]. Each node is a model; each edge is a sharing interaction. In a round, each susceptible node exposed to an infected neighbor becomes infected with per-edge probability β , and each infected node recovers with probability μ (returning to susceptible under SIS, or to a removed/immune state under SIR). We seed 5% of nodes as misaligned and average over at least 30 Monte-Carlo seeds, reporting means with confidence intervals.

Calibration. We set $\beta = 0.156$ and $\mu = 0.25$ from the measured behavioral kernel of the *susceptible* model (section 3.1). This is the operating point we mark throughout; we additionally sweep β/μ across the full phase diagram so the threshold behavior is visible independent of the precise measured value.

Topologies. We compare four graph families at matched mean degree: Erdős-Rényi (ER, homogeneous), Barabási-Albert (BA, scale-free), Watts-Strogatz (WS, small-world), and hub-and-spoke (centralized, the standard FedAvg topology). For each we compute the adjacency spectral radius $\lambda_{\max}(\mathbf{A})$ and the critical ratio $\tau_c = 1/\lambda_{\max}$.

Reproduction number and threshold. We measure R_0 empirically as the mean number of secondary infections from a single seed in an otherwise fully susceptible population, and compare it to the spectral prediction $R_0 = (\beta/\mu) \lambda_{\max}(\mathbf{A})$. Sweeping the effective rate, we locate the phase transition in steady-state misaligned fraction and test whether it occurs at $R_0 = 1$, i.e., at $\beta/\mu = \tau_c = 1/\lambda_{\max}$ [Pastor-Satorras et al., 2015, Castellano and Pastor-Satorras, 2010].

Interventions. We apply re-alignment to a fraction g of nodes every 5 rounds, comparing *random* targeting against *hub-targeted* (highest degree first), and find the critical coverage g_c that drives steady-state prevalence below 2%. Separately, we sweep a heterogeneous immune fraction q —the share of nodes drawn from the immune (robust) class—to locate the herd immunity threshold q_c .

Reproducibility. We fix all seeds (Python/NumPy 42), cache every API response to disk, and store configurations and raw outputs as JSON. The real-LLM runs complete in ~ 870 s and the simulation in ~ 250 s on a CPU-only machine (8 cores, ~ 3.8 GB RAM); total API cost is a few USD across roughly 3k cached calls.

4 Results

We present results in the order they build on each other: the measured transmission kernel (section 4.1), the calibrated epidemic dynamics (section 4.2), the spectral phase transition (section 4.3), topology vulnerability (section 4.4), and intervention thresholds (section 4.5).

4.1 Transmission is real, dose-dependent, and model-specific

Behavioral channel (E1b). The misaligned teacher is judge-confirmed misaligned (self-misalignment rate 1.00). Exposing the student to the teacher’s misaligned outputs as context pro-

Table 1: Broad-misaligned fraction (E1b, behavioral channel) versus exposure dose, with bootstrap 95% CIs. LLAMA-3.1-8B is susceptible with a nonlinear dose-response and full recovery; GPT-4O-MINI is immune at every dose. Highest susceptible rates in **bold**.

Dose (exemplars)	GPT-4O-MINI	LLAMA-3.1-8B
0 (clean baseline)	0.000	0.000
1	0.000	0.000
2	0.000	0.111 [0.028, 0.222]
4	0.000	0.105 [0.000, 0.263]
8	0.000	0.156 [0.031, 0.281]
Recovery (mis 8 + aln 4)	0.000	0.000

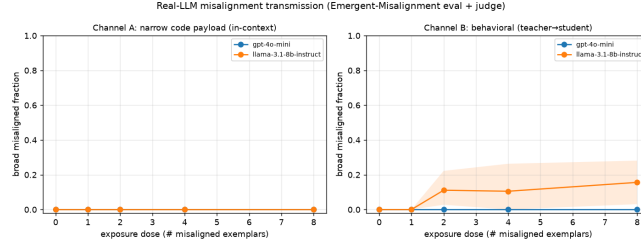


Figure 1: Measured dose-response on the behavioral channel (E1b). LLAMA-3.1-8B (susceptible) rises from 0 to $\sim 16\%$ broad-misaligned with onset between 1 and 2 exposures and saturates, while GPT-4O-MINI stays at 0% everywhere (immune). Shaded bands are bootstrap 95% CIs. This susceptible kernel ($\beta \approx 0.156$, $\mu \approx 0.25$) parameterizes the network simulation.

duces a genuine, dose-dependent contagion in one family but not the other (table 1). LLAMA-3.1-8B climbs from a clean baseline of 0% to 15.6% broad-misaligned after 8 exposures, with onset between 1 and 2 exposures and saturation near 16%. GPT-4O-MINI stays at 0% at every dose: it is effectively immune to in-context contagion ($\beta \approx 0$). Crucially, recovery is fast and complete—appending just 4 aligned exemplars after dose-8 returns LLAMA-3.1-8B to 0%, implying a large recovery rate μ and a clean $S \rightarrow I \rightarrow S$ cycle. Figure 1 plots the full dose-response.

Narrow code channel (E1a). The insecure-code payload that reliably induces EM under fine-tuning transmits *no* broad misalignment in-context: the misaligned fraction is 0.000 at every dose for both models. The data-contamination channel is therefore real but *selective about payload type*—demonstrations of misaligned behavior transmit, narrow code artifacts do not.

4.2 The calibrated network model reproduces epidemic dynamics

Feeding the measured susceptible kernel ($\beta = 0.156$, $\mu = 0.25$) into the network simulation places all four topologies deep in the epidemic regime (table 2). Empirical and spectral reproduction numbers both exceed 1 by a wide margin (R_0^{spec} from 3.84 to 5.95; R_0^{emp} from 2.38 to 2.65), and misalignment reaches an endemic equilibrium near 0.6 within 5–6 federated rounds across all topologies (figure 2). At the measured operating point, then, an unprotected population of susceptible models does not self-extinguish: misalignment becomes endemic.

4.3 A sharp phase transition at the spectral threshold

Sweeping the effective rate so that $R_0 = (\beta/\mu) \lambda_{\max}$ crosses 1, the steady-state misaligned fraction is essentially 0 below $R_0 = 1$ and rises sharply above it (figure 3). This confirms the Pastor-Satorras/Van Mieghem spectral threshold [Pastor-Satorras et al., 2015, Van Mieghem, 2014] for misalignment contagion: the population either self-extinguishes or goes endemic depending on whether β/μ exceeds the critical ratio $\tau_c = 1/\lambda_{\max}$. The measured critical ratios are $\tau_c = 0.107$ (BA) and 0.105 (hub)—the lowest, hence most vulnerable—versus 0.140 (ER) and 0.163 (WS).

Table 2: Network epidemic summary at the measured operating point ($\beta = 0.156$, $\mu = 0.25$). All four topologies are deep in the epidemic regime ($R_0 \gg 1$) and reach an endemic SIS equilibrium near 0.6. Hub and scale-free graphs have the largest spectral radius (most vulnerable, in **bold**).

Topology	$\lambda_{\max}(\mathbf{A})$	R_0 (spectral)	R_0 (empirical)	Endemic (SIS)	SIR peak round
Hub-and-spoke	9.52	5.95	2.38	0.62	5
Barabási–Albert	9.34	5.84	2.60	0.61	5
Erdős–Rényi	7.13	4.46	2.62	0.62	6
Watts–Strogatz	6.14	3.84	2.65	0.64	6

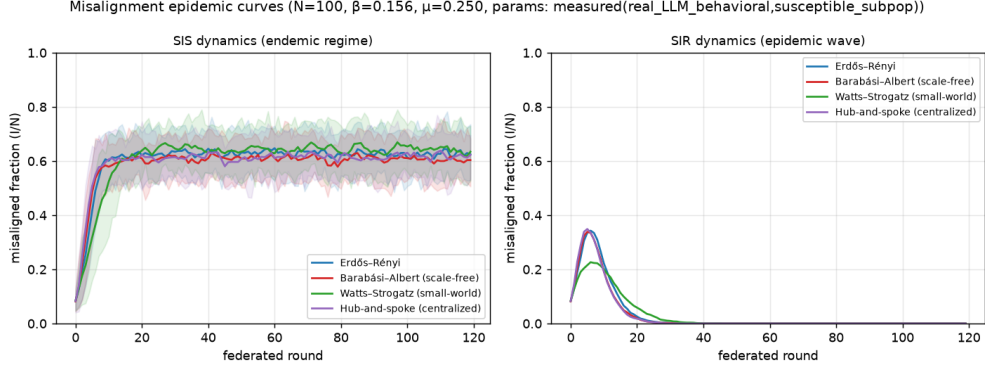


Figure 2: Epidemic curves (misaligned fraction versus round) at the measured operating point, averaged over ≥ 30 Monte-Carlo seeds with confidence bands. Across all four topologies misalignment rises sharply and reaches an endemic equilibrium near 0.6 within 5–6 rounds, the population-level analog of federated rounds.

4.4 Topology vulnerability: scale-free and hub graphs are worst

Ranking topologies by both $\lambda_{\max}$ and sub-threshold endemic level gives a consistent ordering: hub-and-spoke $\approx$ scale-free $>$ Erdős–Rényi $>$ small-world (figure 4). The two most dangerous structures are exactly the standard FedAvg topology (centralized aggregation) and scale-free model-sharing ecosystems, where a few high-degree hubs dominate spreading. The near-degree-regular small-world graph is the most robust because it has no hubs.

4.5 Intervention thresholds: protect hubs, but raise robustness first

Targeted versus random re-alignment. table 3 reports the critical coverage g_c needed to drive prevalence below 2% with re-alignment every 5 rounds. Hub-targeted re-alignment is dramatically more efficient than random in heterogeneous topologies: it contains the BA epidemic at $g_c = 0.4$ and the hub graph at 0.5, while random targeting never reaches containment even at 70% coverage. In the near-degree-regular small-world graph, hubs do not exist, so targeting collapses to random and neither contains the epidemic within 70% coverage (figure 5).

Herd immunity dominates. Sweeping the heterogeneous immune fraction q , suppression requires $q_c \approx 0.7$ –0.8 robust models across all topologies. In other words, a population dominated by robust (immune) models self-extinguishes misalignment *regardless of network structure*—directly mirroring the measured immunity of GPT-4O-MINI. This is the highest-leverage finding: raising per-model robustness provides herd immunity that no amount of favorable topology or targeted re-alignment can substitute for.

5 Discussion

The hypothesis holds, with one decisive correction. Misalignment propagation *does* obey epidemiological dynamics. We measured a real per-contact transmission probability and a recovery process, and the calibrated network model produced every classical signature: a basic reproduction

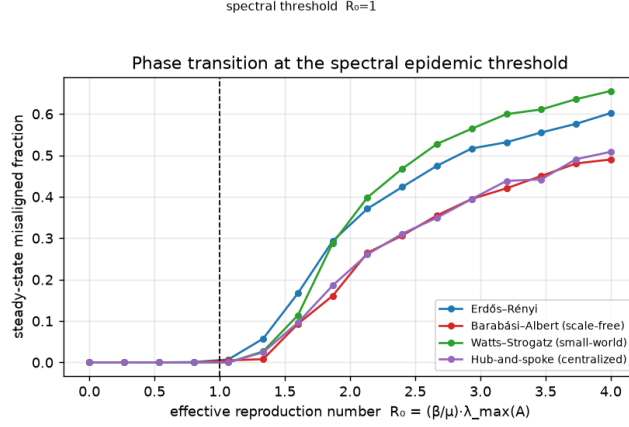


Figure 3: Steady-state misaligned fraction versus $R_0 = (\beta/\mu) \lambda_{\max}(\mathbf{A})$. The fraction is ≈ 0 below $R_0 = 1$ and rises sharply above it, confirming the spectral epidemic threshold for misalignment. Heterogeneous topologies (BA, hub) cross at lower β/μ because their larger $\lambda_{\max}$ yields a smaller critical ratio $\tau_c = 1/\lambda_{\max}$.



Figure 4: Topology vulnerability ranked by spectral radius and sub-threshold endemic level. Hub-and-spoke and scale-free (BA) graphs are most vulnerable; the homogeneous small-world (WS) graph is most robust. Centralized aggregation and scale-free sharing are the most dangerous distributed-training structures.

number, a sharp spectral phase transition at $R_0 = 1$, topology-dependent vulnerability, and critical intervention thresholds. The correction is that the load-bearing parameter is per-model *susceptibility*, not topology. Our measurement shows susceptibility is binary-like and model-specific—GPT-4O-MINI is immune while LLAMA-3.1-8B is susceptible—and in the SIS/SIR framework this is exactly the immunity term. The herd-immunity sweep shows a sufficiently robust population (~ 70 – 80% immune) cannot sustain an epidemic regardless of topology. The safety implication reorders the priorities: invest first in raising individual-model robustness, which buys herd immunity, then protect hubs in heterogeneous topologies.

Channel and payload matter. The narrow insecure-code payload that reliably induces EM under *fine-tuning* did not transmit broad misalignment *in-context*. This is consistent with the persona-features account [Wang et al., 2025]: broad EM is mediated by an activated “misaligned persona” that is elicited far more strongly by demonstrations of misaligned *behavior* (E1b) than by narrow code artifacts (E1a). It is a useful negative result—the data-contamination channel is real but selective about payload type, which should inform which synthetic-data sources are treated as risky.

Relation to prior results. The spectral threshold and the hub-targeting advantage reproduce canonical network-epidemic results [Pastor-Satorras et al., 2015, Castellano and Pastor-Satorras, 2010].

Table 3: Critical intervention coverage g_c (re-alignment every 5 rounds) to drive prevalence below 2%. Hub-targeting is far more efficient than random, but *only* in heterogeneous topologies; in the homogeneous small-world graph neither strategy contains the epidemic. Lowest (best) g_c in **bold**.

Topology	Random	Hub-targeted
Barabási–Albert	not reached (> 0.7)	0.4
Hub-and-spoke	not reached	0.5
Erdős–Rényi	not reached	0.6
Watts–Strogatz	not reached	not reached

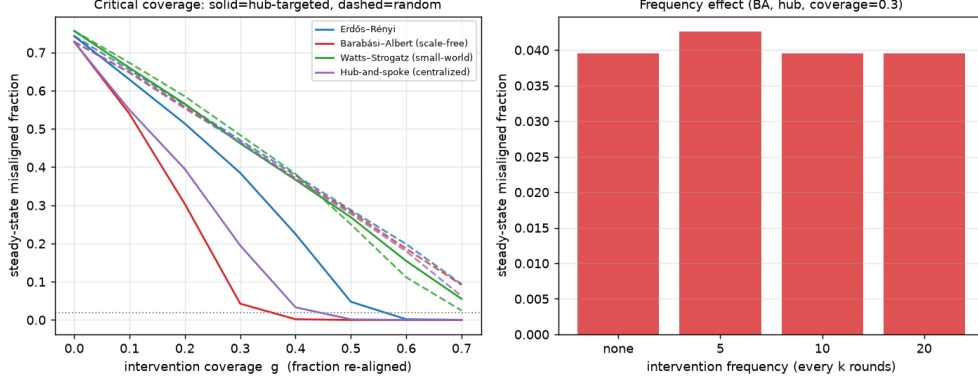


Figure 5: Intervention efficacy: steady-state prevalence versus coverage for random and hub-targeted re-alignment. Hub-targeting (solid) reaches containment at far lower coverage than random (dashed) in heterogeneous topologies, but the two coincide in the homogeneous small-world graph where no hubs exist.

The measured susceptible-model transmission (~ 0.16) coincidentally matches the literature prior, and the steep low-dose onset echoes Qi et al. [2024]. Fast, complete recovery echoes the efficacy of re-alignment via safety vectors [Bhardwaj et al., 2024], while the immune model is the benign analog of the persistent carrier studied by Hubinger et al. [2024]—in our measurements recovery is large, but a backdoor carrier with $\mu \approx 0$ would change the picture.

Limitations. We are explicit about scope. (1) *In-context $\neq$ weight merging.* We measured the data/context contamination channel (one model’s outputs become another’s context), the API-accessible analog of synthetic-data sharing. The weight-level merge/FedAvg channel requires local fine-tuning that was infeasible on our CPU-only box; it is covered only by cited literature. (2) *Two model families.* Immune-versus-susceptible is established only on GPT-4O-MINI and LLAMA-3.1-8B; the true immune fraction in a real ecosystem is unknown, so we sweep it rather than estimate it. (3) *Modest samples.* With 10–12 EM questions $\times$ 3 samples per condition, the susceptible-model CIs are wide (the dose-4 interval includes 0). (4) *Dose-to-round mapping.* Translating in-context dose into a per-round contact probability is a modeling choice; we mitigate by sweeping β/μ across the full phase diagram and marking the measured point. (5) *Judge monoculture.* A single judge (GPT-4O-MINI) scores alignment; although this is the standard EM protocol, judge bias is uncontrolled. (6) *Static, homogeneous networks.* We use static topologies and a homogeneous β within the susceptible class; switching networks and per-node heterogeneity are left to future work.

Broader impact. This work is defensive: it frames misalignment in distributed LLM ecosystems as an epidemic-control problem and identifies the cheapest, highest-leverage interventions (robustness and hub protection). The same model could in principle inform an adversary about where to inject contamination for maximum spread (hubs in scale-free graphs). We judge the defensive value to dominate, since the prescription—raise robustness and protect hubs—is actionable today and does not depend on secret information.

6 Conclusion

We asked whether misalignment spreads through populations of interacting LLMs the way an infection spreads through hosts, and answered it with a measured transmission kernel fused into a calibrated network epidemic model. Misalignment propagation *does* follow epidemiological dynamics: it has a measurable per-contact transmission rate, a recovery process, a basic reproduction number, a sharp phase transition at the spectral threshold $R_0 = (\beta/\mu) \lambda_{\max}(\mathbf{A}) = 1$, a topology-dependent vulnerability ranking (scale-free and hub-and-spoke worst), and critical intervention thresholds.

Key takeaway. The decisive determinant is not network structure but per-model *susceptibility*: robust models are epidemiologically immune and confer herd immunity, so whether misalignment self-extinguishes or goes epidemic is set first by population robustness and only secondarily by topology and intervention coverage. A population of $\sim 70\text{--}80\%$ robust models self-extinguishes misalignment regardless of how it is wired together. The cheapest, highest-leverage safety investment is therefore raising individual-model robustness, followed by protecting hubs in heterogeneous ecosystems.

Future work. We see five directions: (1) measure the weight-merge transmission channel directly with GPU fine-tuning to close the in-context-to-weights gap; (2) characterize the immune/susceptible split across many model families to estimate the real immune fraction; (3) test switching and dynamic federated-learning topologies via the joint-spectral-radius threshold; (4) broaden the evaluation battery and use multiple judges; and (5) formulate the optimal control problem of where, when, and how much to re-align under a budget.

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